Experiment No:3

Student Name: Sumanyu Singh
Branch: BE CSE
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UID:23BCS10801
Section/Group:KRG-3B
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Aim: To design and implement a secure web navigation mechanism using context-based authentication, protected routing techniques, and optimized React hooks to enhance application performance along with an improved Material UI interface.

Objective:

- To handle authentication status across the entire application using the Context API
- To enable efficient navigation between multiple pages using routing mechanisms
- To prevent unauthorized users from accessing restricted pages
- To allow navigation only after successful user authentication
- To automatically redirect unauthenticated users to the login interface
- To strengthen application security while improving user interaction
- To develop a modular, reusable, and maintainable code architecture

Implementation/Code:

1. **Log.js:-**

```
export const logs = [
  { id: 1, activity: "Car Travel", carbon: 4 },
  { id: 2, activity: "Electricity Usage", carbon: 6 },
  { id: 3, activity: "Cycling", carbon: 0 },
];
```

2. **Dashboardlayout.jsx:-**

```
jes > DashboardLayout.jsx > DashboardLayout
import {Link, Outlet} from 'react-router-dom';

const DashboardLayout=()=>{
    return(
        <>
        <h3>DashBoard</h3>

        <nav>
            <Link to="summary">Summary</Link> |{" "}
            <Link to="Analytics"></Link> |{" "}
        </nav>
        <hr />
        <Outlet />
        </>
    )
}
export default DashboardLayout;
```

3. **summary.jsx:**

```
const DashBoardSummary=()=>{
    return(
        <div>
            <h3>Summary</h3>
            <p>Summary content goes here.</p>
        </div>
    )
}
export default DashBoardSummary;
```

4. **App.jsx:**

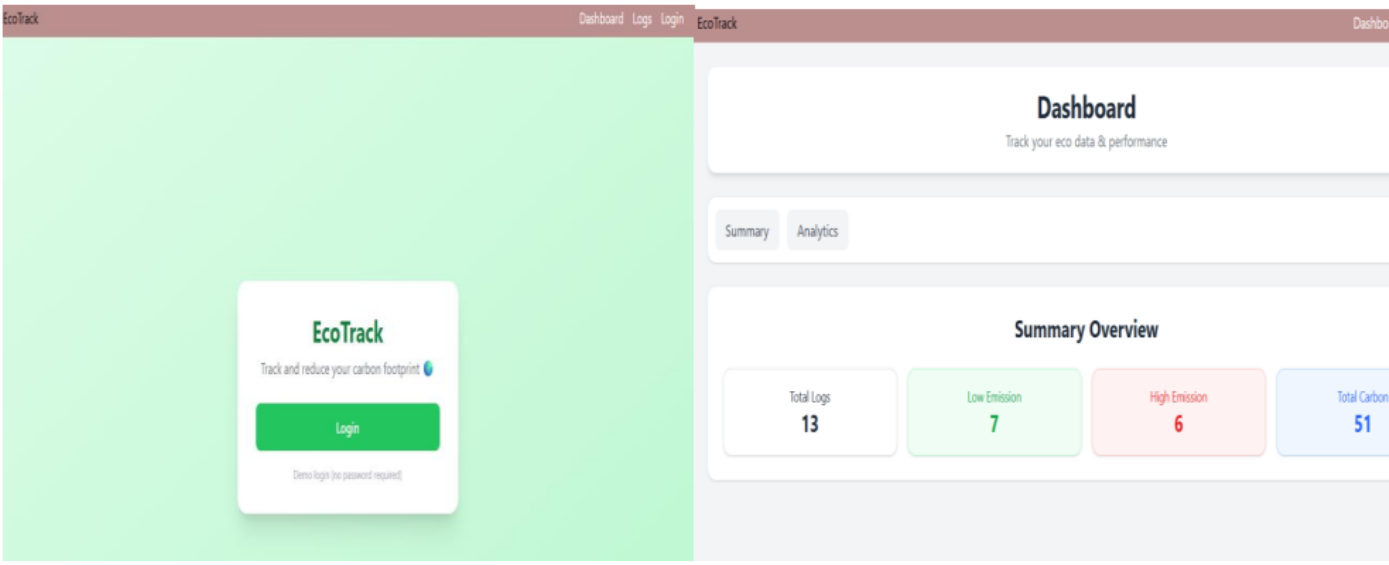
```
pp.jsx > ...
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Header from "./components/Header";
import Login from "./pages/Login";
import DashboardLayout from "./pages/DashboardLayout";
import ProtectedRoute from "./routes/ProtectedPoute";
import DashBoardSummary from "./pages/DashboardSummary";
import DashBoardAnalytics from "./pages/DashboardAnalytics";

const App = () => {
  return (
    <BrowserRouter>
      <Header />

      <Routes>

        <Route path="/login" element={<Login />} />


        <Route
          path="/"
          element={
            <ProtectedRoute>
              <DashboardLayout />
            </ProtectedRoute>
          }
        >
          <Route index element={<DashBoardSummary />} />
          <Route path="summary" element={<DashBoardSummary />} />
          <Route path="analytics" element={<DashBoardAnalytics />} />
        </Route>
      </Routes>
    </BrowserRouter>
  );
};

export default App;
```

5. **Authcontext.js**

context > AuthContext.jsx > ...

```
import { createContext, useContext, useState } from "react";
//sumanyu singh code

const AuthContext = createContext(null);

export const AuthProvider = ({ children }) => {
    const [isAuthenticated, setIsAuthenticated] = useState(false);

    return (
        <AuthContext.Provider value={{ isAuthenticated, setIsAuthenticated }}>
            {children}
        </AuthContext.Provider>
    );
};

export const useAuth = () => useContext(AuthContext);
```

OUTPUT:



Learning Outcome :-

- Gained understanding of performance optimization using advanced React hooks
- Learned implementation of protected routing using Context API
- Reduced unnecessary component updates and improved efficiency
- Developed modern user interfaces using Material UI
- Enhanced application scalability, readability, and long-term maintainability